From Raw FIT files to Explainable AI:

Reconstructing real training behavior from raw data — and linking it to physiological adaptation

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Context

In endurance sports, large volumes of high-resolution data are continuously collected. At the highest level, this data is often combined with laboratory testing and expert interpretation. Physiological parameters can be measured, monitored, and translated into targeted training decisions. However, this process remains resource-intensive, requiring time, expertise, controlled testing environments and interpretation. For many athletes — even those training seriously — this level of precision isn't accessible more often than not. At the same time, there is a clear demand to better understand not just what to do, but why it works.

This work originates from a simple observation: athletes speak in intervals.

These are not just numbers, but shared, immediately understood building blocks of training. Both athlete and coach operate within this language: the athlete through experience, the coach through the physiological structure behind it.

This project aims to extract training structures directly from executed sessions and express them as a distribution of these conventional interval types — grounding the analysis in a language that is already used in practice, while providing more specificity than broad training zones alone.

What I built

I developed a **data pipeline** that reconstructs real training behaviour from raw FIT files, focusing on signal quality, physiological consistency, and interpretability. **Core components** include:

- Parsing and structuring large volumes of raw training data (800+ FIT files)
- Smoothing and denoising high-resolution time series (1 Hz data)
- Detecting intervals based on physiologically grounded thresholds
- Reconstructing sessions as a distribution of conventional interval types
- Aggregating session-level features linked to adaptation metrics (e.g. Efficiency Factor, aerobic decoupling)
- Integrating contextual information, including cross-training (v1.1)
- Applying machine learning models (Random Forest) to relate observed training behaviour to adaptation outcomes

The focus throughout is not prediction accuracy alone, but maintaining a clear and interpretable link between observed training behaviour and un**derlying physiological processes.**

Key insight

Coaching and training in endurance sports is built on a solid set of physiological principles. However, in practice, the growing amount of knowledge and available metrics can easily lead to overcomplication.

The model acts as a way to simplify — not by reducing information, but by structuring it.

It does not replace expertise, but supports it by filtering noise and helping to stay focused on what matters within a given context.
Rather than trying to apply everything at once, it helps maintain direction.

Direction

The next step is to further refine and extend the model. This includes:

- Improving its ability to autonomously detect physiologically relevant patterns from training data, reducing reliance on predefined assumptions.
- Expanding the approach beyond metabolic signals by integrating biomechanical parameters, moving towards a more complete, individual athlete profile.
- Finally, the goal is to operationalize the pipeline into a scalable, automated, cloud-based solution, enabling continuous data ingestion, processing, and analysis.

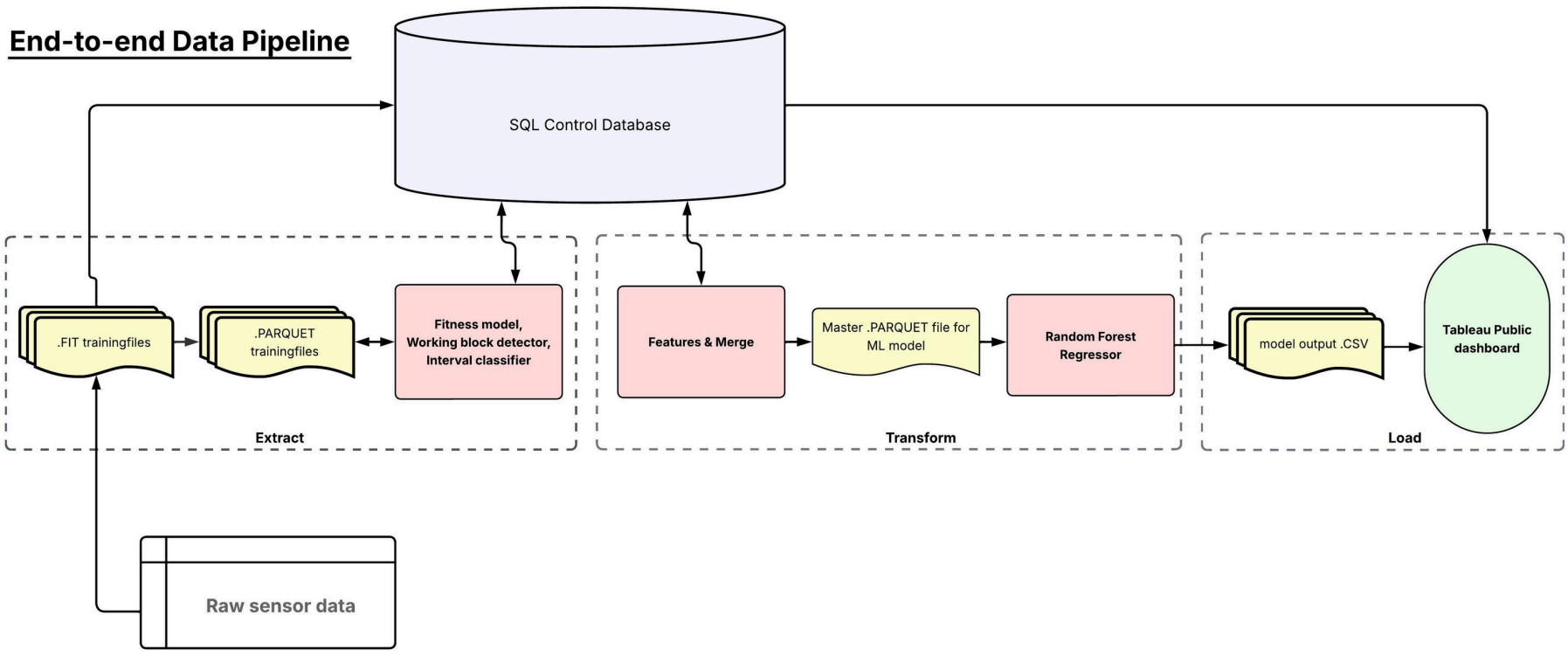
Ideally, this would be explored in an environment where decisions need to be both precise and grounded in real-world training behaviour.

Notes

This work represents an early-stage exploration (v1.1), focused on building a coherent and explainable framework rather than a finished product.

Github / Code

https://github.com/kazverstraete-a11y/cycling-performance-model


End-to-end Data Pipeline
SQL Control Database
.FIT trainingfiles
.PARQUET trainingfiles
Fitness model, Working block detector, Interval classifier
Extract
Features & Merge
Master .PARQUET file for ML model
Random Forest Regressor
Transform
model output .CSV
Tableau Public dashboard
Load
Raw sensor data